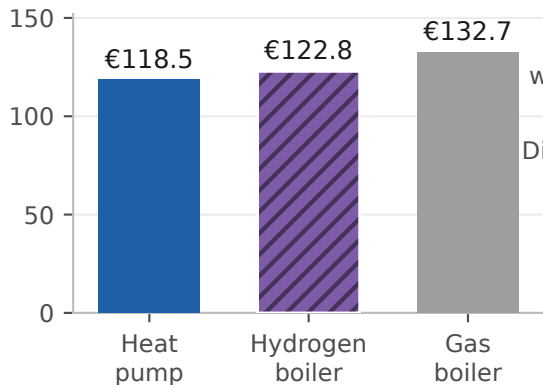


Heat pumps decarbonise Europe's homes, whereas hydrogen is a support-contingent capacity option

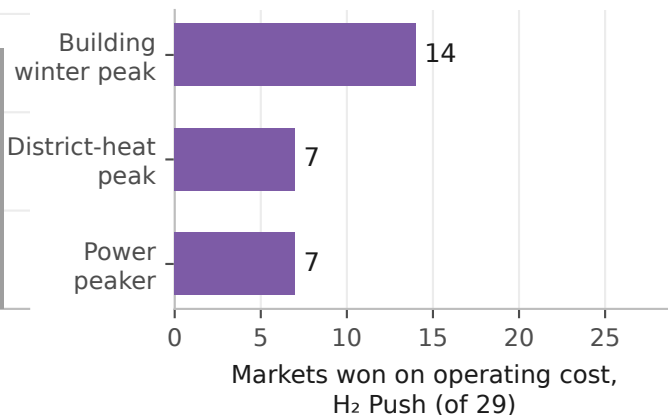
29 European markets (EU-27, UK, Switzerland) · 1,369 NUTS3 regions · 2025 to 2050

2050 levelised cost of heat (€/MWh)

Base load: 2050 delivered cost



Peaking: dispatch wins



Heat pumps beat the hydrogen boiler in 22 of 29 countries, and 5 of the 7 exceptions are salt-cavern markets.

Counts charge both carriers symmetrically, hydrogen paying its own last-mile network. No market recovers the peaker's capital from market rent. On the high carbon path hydrogen wins the power peak only in the 7 salt-cavern markets, a different seven; the 5 that build would need €31 to 63/kW-yr to be financeable.